Effect of Irrigation Saline Temperature on Post-Operative Periorbital Bruising Following Eyelid Surgery: A Randomized Blinded Trial

Contents

1 Methods **5**

1.1 Study Design and Participants 5

1.2 Image Acquisition and Grading 5

1.3 Outcome Measures 5

1.4 Inter-Rater Reliability 6

1.5 Statistical Analysis 6

2 Results **9**

2.1 Descriptive Statistics 9

2.2 Inter-Rater Reliability 9

2.3 Primary Analysis: Linear Mixed Model 9

2.4 Population-Averaged Analysis: GEE-Ordinal 13

2.5 Binary Outcome Analysis 14

2.6 Composite Outcome 14

2.7 Ordinal Analysis: Cumulative Link Mixed Model 15

2.8 Summary Comparison Across Statistical Models 15

2.9 Effect Sizes 15
2.10 Power Analysis 17
2.11 Multi-Model Ensemble and Model Limitations 17
2.12 Sensitivity Analyses 18
2.13 Model Diagnostics 19

3 Discussion **20**
3.1 Main Findings 20
3.2 Consistency Across Model Families 20
3.3 Inadequate Statistical Power 20
3.4 Between-Patient Heterogeneity and Clustering 21
3.5 Inter-Rater Reliability 21
3.6 Procedure Code as a Dominant Predictor 22
3.7 Composite Outcome and Internal Consistency 22

4 Recommendations for Future Paired-Design Trials **23**
4.1 Explicit Eye-Level Treatment Assignment in Raw Dataset 23
4.2 Document Quadrant-to-Eye Mapping Explicitly 23
4.3 Prospectively Document All Exclusion Reasons 24

5 Conclusions **25**

Abstract

Background. Post-operative periorbital bruising is a frequent concern following eyelid surgery, with patient-visible sequelae that may influence recovery experience and satisfaction. Intraoperative irrigation with saline at different temperatures has been proposed as a simple means to reduce local bruising; however, rigorous evidence from controlled trials is lacking. This study aimed to evaluate the effect of irrigation saline temperature on post-operative periorbital bruising using a randomized, grader-blinded design.

Methods. Thirty-four patients undergoing eyelid surgery were enrolled; two were excluded (IDs 19 and 25), yielding 32 analysed patients (Condition 0: $n = 14$; Condition 1: $n = 18$). Post-operative bruising was assessed from standardised photographs by two independent, blinded graders across eight periorbital quadrants per patient (512 total observations). Outcomes were bruising depth and extent, each rated on an ordinal 0–4 scale. Inter-rater reliability was quantified by ICC(2,1) and quadratic-weighted Cohen's kappa with a pre-specified acceptability threshold of ≥ 0.75. Treatment effects were estimated using a multi-model approach: linear mixed models (LMM), cumulative link mixed models (CLMM), and generalised estimating equations (GEE-Ordinal and GEE-Binary), all adjusted for procedure code and grader with a patient-level random intercept.

Results. No statistically significant treatment effect was detected for any outcome across any model. LMM estimates for the condition effect were: depth $\beta = -0.034$ (95% CI: -0.487 to 0.420, $p = 0.885$) and extent $\beta = 0.029$ (95% CI: -0.258 to 0.316, $p = 0.844$). CLMM estimates on the proportional odds scale were: depth OR = 0.822 ($p = 0.702$) and extent OR = 1.006 ($p = 0.987$). All effect sizes were negligible (Cohen's $d < 0.12$; Cliff's $\delta < 0.09$). Inter-rater reliability met the acceptability threshold for extent (ICC = 0.888) but not for depth (ICC = 0.515), with a systematic positive bias from Grader 1 (mean difference +0.184, $p < 0.001$). Post-hoc power was 5.5–7.2%, indicating the trial was severely underpowered; minimum detectable effects were approximately 1.0 SD, requiring 571–2,786 patients per group to achieve 80% power.

Conclusions. No meaningful difference in post-operative periorbital bruising was

observed between treatment conditions across all statistical models and outcome definitions. The findings were consistent and robust to the analytic approach. However, the study was severely underpowered, and the absence of eye-level treatment assignment precluded recovery of the intended paired structure. Future trials should adopt eye-level randomisation with prospective recording of quadrant-to-eye mappings and pre-specified sample size calculations. The study remains blinded at the time of this analysis.

1 Methods

1.1 Study Design and Participants

This was a randomized, grader-blinded clinical trial evaluating the effect of irrigation saline temperature on post-operative periorbital bruising following eyelid surgery. Thirty-four patients were enrolled across three surgical procedure types. Two patients (IDs 19 and 25) were absent from the dataset and excluded from all analyses, yielding a final analytic sample of 32 patients. Treatment was assigned at the patient level: 14 patients were allocated to Condition 0 and 18 patients to Condition 1. Three procedure types were included: procedure code 1 (7 patients in Condition 0 vs. 8 in Condition 1), procedure code 2 (1 vs. 3), and procedure code 3 (6 vs. 7). Reasons for the exclusion of patients 19 and 25 were not available in the dataset.

1.2 Image Acquisition and Grading

Eight standardised photographs were obtained for each patient, capturing four periorbital quadrants per eye (quadrants a–h). Two independent graders, blinded to patient identity and treatment allocation, assessed each image independently. Each image was scored on two dimensions: bruising depth (severity) and bruising extent (area), both on a discrete ordinal scale of 0–4. The complete dataset comprised $32 \times 8 \times 2 = 512$ graded observations.

1.3 Outcome Measures

Primary outcomes. Bruising depth and bruising extent, each on a 0–4 ordinal scale (0 = none, 4 = maximal).

Secondary outcomes. A composite score (depth + extent, range 0–8), and binary indicators of any bruising (≥ 1) and clinically significant bruising (≥ 2), derived separately for depth and extent.

1.4 Inter-Rater Reliability

Inter-rater reliability was quantified using the intraclass correlation coefficient ICC(2,1) for absolute agreement under a two-way mixed-effects model, and quadratic-weighted Cohen's kappa to account for ordinal score proximity. Systematic grader bias was assessed using a paired t-test and a Wilcoxon signed-rank test on within-image score differences (Grader 1 minus Grader 2). The pre-specified acceptability threshold for ICC was ≥ 0.75.

1.5 Statistical Analysis

The primary outcome data are ordinal (0–4), and the dataset contains multiple levels of non-independence: two graders score each image, eight quadrant images are taken per patient, two eyes are photographed per patient, and the intervention is assigned at the patient level. No single model addresses all of these features simultaneously. A multi-model strategy was therefore adopted to examine whether the condition effect estimate was robust across model assumptions, scales of inference, and treatment of the ordinal response.

A Linear mixed model (LMM) was utilized as the primary model treated ordinal scores as approximately continuous and estimated:

$$Y_{ijkl} = \mu + \alpha_k + \pi_l + \gamma_j + b_i + \varepsilon_{ijkl} \tag{1}$$

where Y_{ijkl} is the outcome for patient i, grader j, condition k, and procedure code l; α_k is the fixed effect of condition; π_l is the fixed effect of procedure code; γ_j is the fixed effect of grader; $b_i \sim \mathcal{N}(0, \sigma_b^2)$ is the patient-level random intercept capturing between-patient heterogeneity; and $\varepsilon_{ijkl} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the residual error. Grader was retained as a fixed effect in all models because of the observed systematic grader bias (see Results).

To respect the ordinal nature of the response, a proportional Cumulative link mixed model (CLMM) was fitted:

$$\Pr(Y_{ijkl} \leq m) = \text{logistic}(\theta_m - \alpha_k - \pi_l - \gamma_j - b_i)\,, \quad m = 0, 1, 2, 3 \tag{2}$$

where θ_m are ordered threshold parameters, $\text{logistic}(x) = 1/(1+e^{-x})$, and all other terms are as in Equation 1. The exponentiated condition coefficient yields a proportional odds ratio (OR): OR < 1 indicates lower cumulative odds (i.e. lower scores) in Condition 1.

To obtain population-average, generalized estimating equations – ordinal (GEE-Ordinal) estimates was implemented to make no distributional assumptions about patient-level heterogeneity, GEE with an exchangeable within-patient correlation structure was fitted:

$$\mathrm{E}[Y_{ij}] = \beta_0 + \beta_1 X_{\text{cond},ij} + \beta_2 X_{\text{proc},ij} + \beta_3 X_{\text{grader},ij} \tag{3}$$

with robust (sandwich) standard errors to guard against misspecification of the working correlation matrix.

For clinical interpretability, binary outcomes (any bruising and clinically significant bruising) were modelled using a generalized estimating equations – binary (GEE-Binary) with logit link and binomial family:

$$\log\left(\frac{p_{ij}}{1-p_{ij}}\right) = \beta_0 + \beta_1 X_{\text{cond},ij} + \beta_2 X_{\text{proc},ij} + \beta_3 X_{\text{grader},ij} \tag{4}$$

Exponentiated coefficients yield odds ratios; absolute risk differences and numbers needed to treat (NNT) or harm (NNH) were computed from marginal predicted probabilities.

A within-patient Conditional logistic model (exploratory) approach was considered to leverage the paired structure (two eyes per patient) for binary outcomes. The conditional likelihood takes the form:

$$L_{\text{cond}}(\boldsymbol{\beta}) = \prod_{i=1}^{n} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_{i,\text{case}})}{\sum_{j \in S_i} \exp(\boldsymbol{\beta}^\top \mathbf{x}_{ij})} \tag{5}$$

where S_i is the stratum of patient i. This model could not be estimated from the current dataset because the treatment variable is constant within each patient stratum (condition is assigned at the patient level, not the eye level), eliminating all within-stratum contrast. It is documented here to motivate the design recommendation for future studies.

Standardised effect sizes were computed on patient-level mean scores (averaged across quadrants and graders) to obtain approximately independent observations at the unit of

randomisation. Cohen's d and Hedges' g (bias-corrected for small samples) were computed from the pooled standard deviation of patient means. Cliff's δ was computed as a non-parametric, ordinal-appropriate effect size. Thresholds for interpretation: negligible $|d| < 0.20$; $|\delta| < 0.147$.

Post-hoc power was estimated for a two-sample t-test at $\alpha = 0.05$ (two-sided) using the observed patient-level Cohen's d and the actual group sizes ($n_0 = 14$, $n_1 = 18$). The minimum detectable effect (MDE) at 80% power was back-calculated, and the required sample size per group for 80% power at each observed effect was also reported.

Five sensitivity domains were examined: (1) *random-effects structure* — OLS (no random effects), random intercept only, random intercept plus random slope, and two-level nested intercepts; (2) *interaction* — condition $\times$ procedure code interaction terms; (3) *grader handling* — averaging scores across graders prior to fitting vs. retaining grader as a fixed effect; (4) *nonparametric* — Mann–Whitney U test, permutation test (10,000 permutations), and Cliff's δ on patient-level means; (5) *multiple comparisons* — Bonferroni and Benjamini–Hochberg corrections applied to depth and extent p-values jointly.

Residuals from the LMM were assessed for approximate normality using the Shapiro–Wilk test and visual inspection of quantile–quantile plots. Variance components (between-patient σ_b^2, residual σ_ε^2) were extracted to quantify the intraclass correlation at the patient level. Scale-location plots and leverage statistics were examined to identify influential observations.

2 Results

2.1 Descriptive Statistics

The analytic sample comprised 32 patients (Condition 0: $n = 14$; Condition 1: $n = 18$) contributing 512 graded observations (32 patients $\times$ 8 quadrant images $\times$ 2 graders). Condition 0 contributed 224 observations and Condition 1 contributed 288 observations. Mean depth scores were numerically similar across conditions, with no clinically meaningful separation apparent from raw score distributions. Score 0 (no bruising) was more frequent in Condition 1 for both outcomes; higher scores (3–4) were uncommon in both groups. The distributions of depth and extent scores by condition are shown in Figure 1 and Figure 2.

Procedure code was identified as a potential confounder: procedure code 3 patients exhibited markedly higher depth and extent scores than those undergoing procedure codes 1 or 2. This is illustrated in Figure 3.

2.2 Inter-Rater Reliability

For bruising depth, ICC(2,1) was 0.515 (95% CI: -0.10 to 0.87), below the pre-specified acceptability threshold of 0.75. For bruising extent, ICC(2,1) was 0.888 (95% CI: 0.55 to 0.98), exceeding the threshold. Grader 1 assigned systematically higher depth scores than Grader 2: mean difference $+0.184$ (SD $= 0.595$), paired t-test $t = 4.93$, $p < 0.001$. Because of this systematic bias, grader was retained as a fixed effect in all primary models rather than being averaged out. The scatter plot of within-image grader agreement for depth is shown in Figure 4; the Bland–Altman plot for extent is shown in Figure 5.

2.3 Primary Analysis: Linear Mixed Model

The LMM (Equation 1) yielded no evidence of a treatment effect for either primary outcome. Full results are presented in Table 1.

For depth, the condition coefficient was $\beta = -0.034$ (SE $= 0.231$; 95% CI: -0.487 to 0.420; $p = 0.885$). For extent, $\beta = 0.029$ (SE $= 0.146$; 95% CI: -0.258 to 0.316;

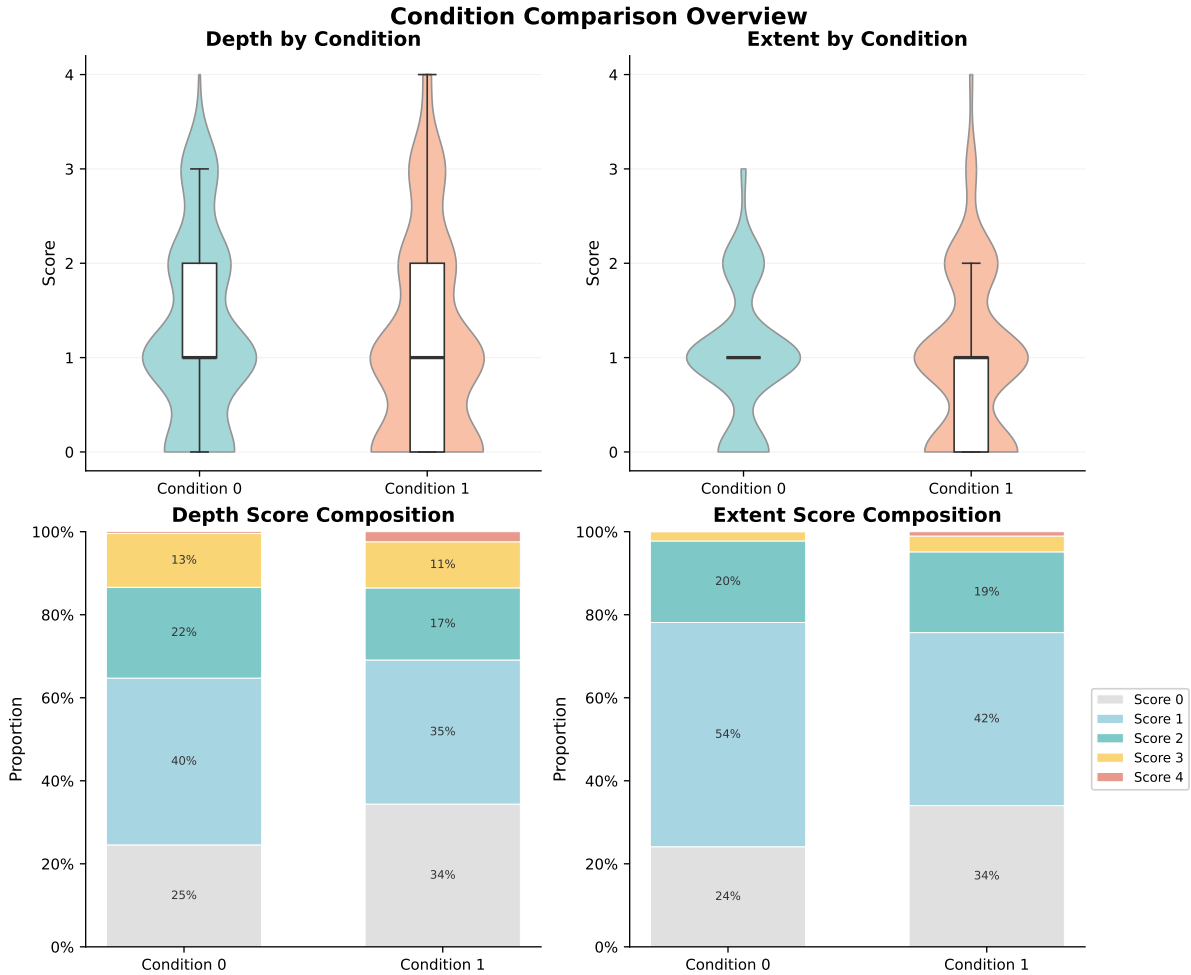

Figure 1: **Comparison of bruising outcomes by treatment condition.** Distribution of depth and extent scores across all 512 observations (32 patients × 8 quadrants × 2 graders) for Condition 0 ($n = 224$ observations) and Condition 1 ($n = 288$ observations). No clinically meaningful difference was observed for either outcome, consistent with the linear mixed model estimates (Table 1).

Table 1: Linear mixed model estimates for the effect of treatment condition on bruising outcomes. Models adjusted for procedure code and grader with a patient-level random intercept. Negative β indicates lower scores in Condition 1 relative to Condition 0.

Outcome	β	SE	95% CI	p-value	Cohen's d
Depth	−0.034	0.231	−0.487 to 0.420	0.885	−0.045
Extent	0.029	0.146	−0.258 to 0.316	0.844	0.044
Composite	−0.005	0.369	−0.728 to 0.719	0.990	−0.004

Note: CI, confidence interval; SE, standard error. All Cohen's d values are classified as negligible ($|d| < 0.20$). $N = 32$ patients, 512 observations.

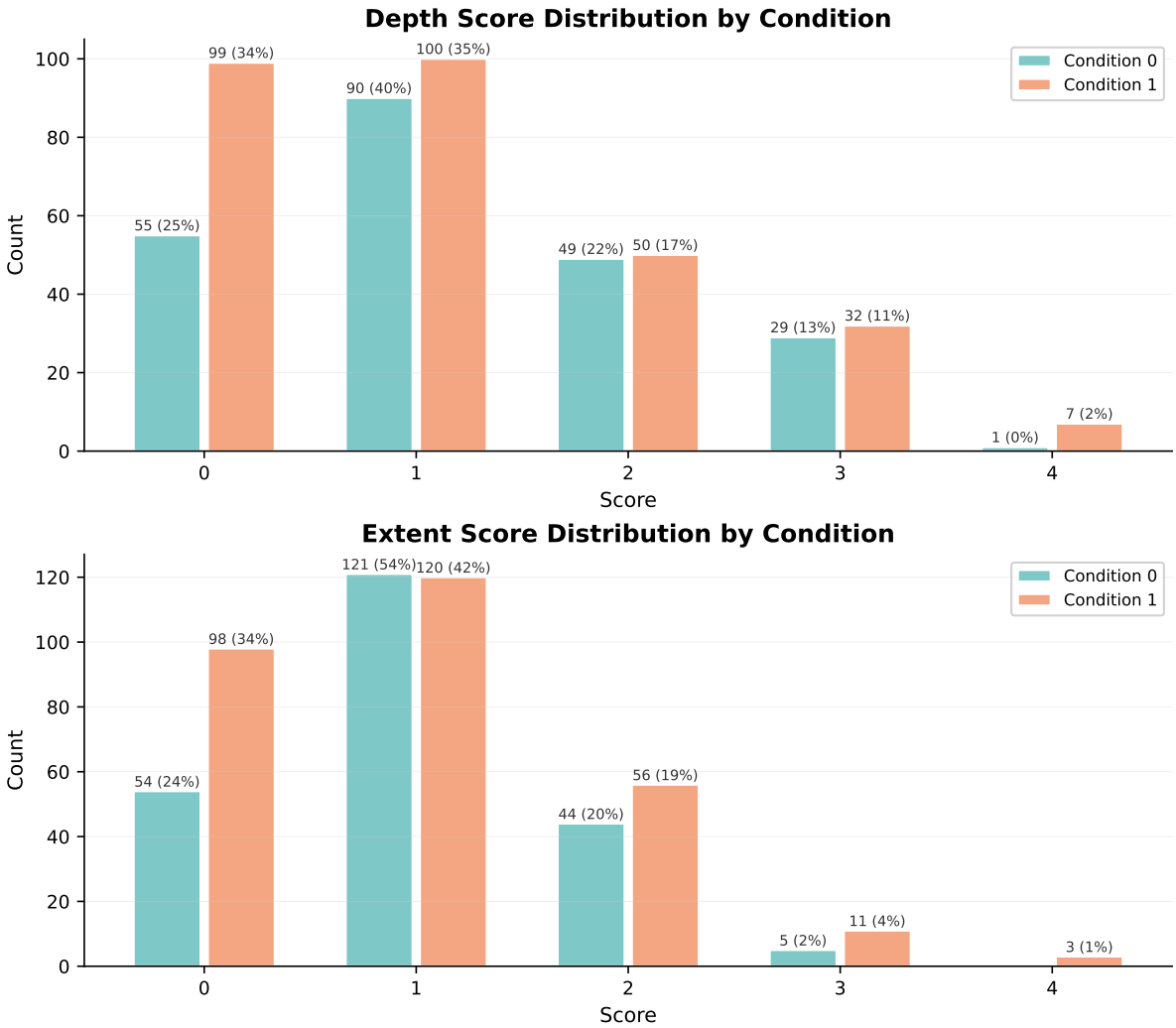

Figure 2: **Score distributions for depth and extent by condition.** Proportion of observations at each ordinal score level (0–4) for Condition 0 and Condition 1. Score 0 was more frequent in Condition 1 for both outcomes; higher scores (3–4) were uncommon in both groups.

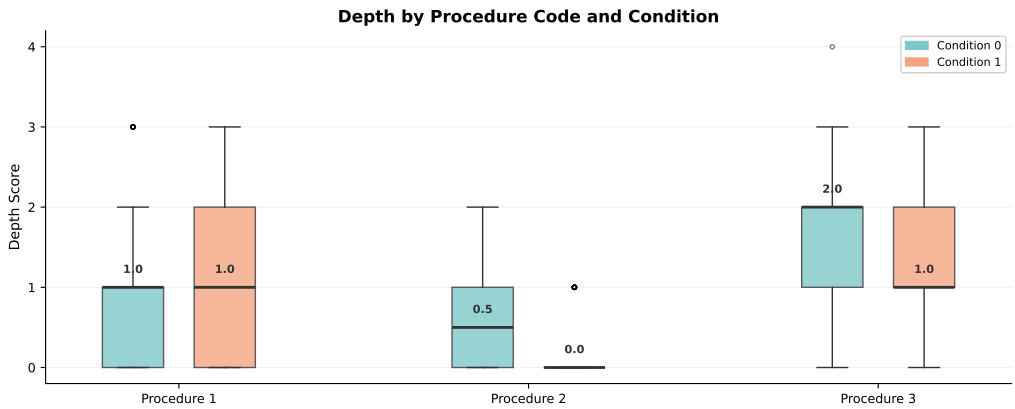

Figure 3: **Bruising depth by procedure code.** Box plots of depth scores stratified by surgical procedure code. Procedure code 3 was associated with markedly higher scores than codes 1 and 2, consistent with the linear mixed model (procedure code 3: $\beta = 0.675$, $p = 0.006$).

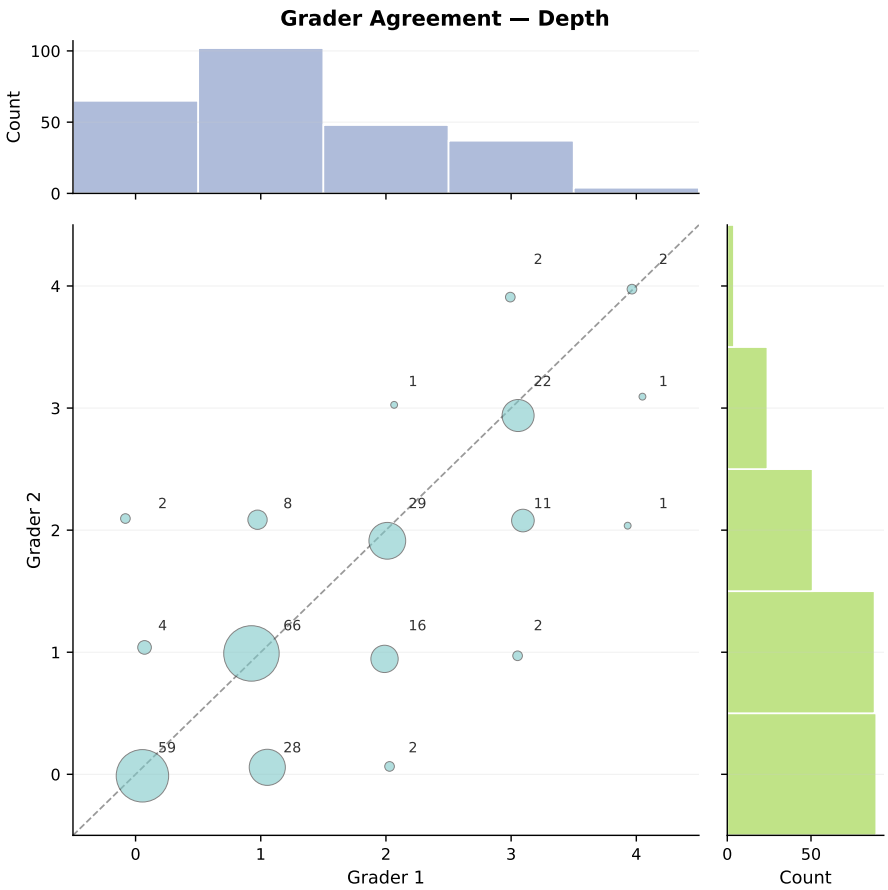

Figure 4: **Inter-rater scatter plot for bruising depth.** Grader 1 scores (y-axis) plotted against Grader 2 scores (x-axis) for the depth outcome across all image pairs. Grader 1 assigned systematically higher scores (mean difference +0.184, $p < 0.001$), reflected in the upward displacement from the identity line. ICC(2,1) = 0.515, below the pre-specified 0.75 acceptability threshold.

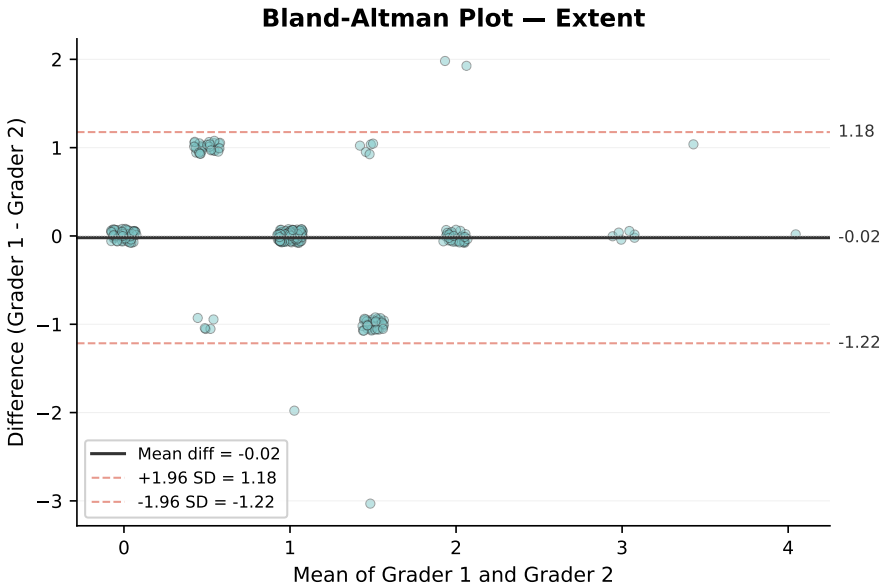

Figure 5: **Bland–Altman plot for bruising extent inter-rater agreement.** Within-image difference (Grader 1 minus Grader 2) plotted against the mean score across all image pairs. The mean difference was −0.020 (dashed line), with limits of agreement shown as dotted lines. Extent ICC(2,1) = 0.888, exceeding the pre-specified acceptability threshold.

$p = 0.844$). The composite outcome similarly showed no effect: $\beta = -0.005$ (SE = 0.369; 95% CI: −0.728 to 0.719; $p = 0.990$).

Procedure code was a significant predictor: procedure code 3 was associated with $\beta = 0.675$ higher depth ($p = 0.006$) and $\beta = 0.571$ higher extent ($p \leq 0.006$) relative to the reference category.

Variance components indicated that $\sigma^2_{\text{patient}} = 0.379$ and $\sigma^2_{\text{residual}} = 0.546$ for depth, yielding a patient-level intraclass correlation of 41.0%. For extent, $\sigma^2_{\text{patient}} = 0.139$ and $\sigma^2_{\text{residual}} = 0.423$ (ICC = 24.7%). These estimates confirm substantial between-patient heterogeneity and justify the inclusion of a random intercept.

2.4 Population-Averaged Analysis: GEE-Ordinal

GEE-Ordinal analysis (Equation 3) produced population-averaged estimates that were nearly identical to the LMM subject-specific estimates: depth $\beta = -0.034$ (SE = 0.214; 95% CI: −0.454 to 0.387; $p = 0.876$) and extent $\beta = 0.029$ (SE = 0.133; 95% CI: −0.232 to 0.290; $p = 0.828$). Results are presented in Table 2.

Table 2: Generalized estimating equations (GEE) estimates for ordinal outcomes with exchangeable within-patient correlation structure. Population-averaged estimates with robust (sandwich) standard errors.

Outcome	β	SE	95% CI	p-value	Cohen's d
Depth	−0.034	0.214	−0.454 to 0.387	0.876	−0.036
Extent	0.029	0.133	−0.232 to 0.290	0.828	0.039

Note: GEE with exchangeable within-patient correlation; robust (sandwich) SE. $N = 32$ patients, 512 observations.

2.5 Binary Outcome Analysis

Binary GEE analyses (Equation 4) for the four dichotomised outcomes are presented in Table 3. None of the binary outcome comparisons reached statistical significance.

Table 3: GEE estimates for binary bruising outcomes. OR < 1 indicates lower odds of bruising in Condition 1. None of the estimates reached statistical significance.

Outcome	Threshold	OR	95% CI	p-value	Risk Diff.	NNT/NNH
Depth	≥ 1	0.703	0.372 to 1.330	0.279	−0.098	10.2
Depth	≥ 2	0.935	0.363 to 2.409	0.889	−0.044	22.9
Extent	≥ 1	0.689	0.363 to 1.305	0.253	−0.099	10.1
Extent	≥ 2	1.256	0.576 to 2.741	0.566	0.024	41.1*

Note: GEE with logit link, binomial family, exchangeable within-patient correlation, and robust SE. *NNH (number needed to harm); point estimate favoured Condition 0 for this threshold. Risk Diff., absolute risk difference. $N = 32$ patients, 512 observations.

For depth ≥ 1 (any bruising), OR = 0.703 (95% CI: 0.372–1.330; $p = 0.279$; risk difference = −0.098; NNT = 10.2). For depth ≥ 2 (clinically significant), OR = 0.935 ($p = 0.889$; risk difference = −0.044; NNT = 22.9). For extent ≥ 1, OR = 0.689 (95% CI: 0.363–1.305; $p = 0.253$; risk difference = −0.099; NNT = 10.1). For extent ≥ 2, OR = 1.256 ($p = 0.566$; risk difference = +0.024; NNH = 41.1).

2.6 Composite Outcome

The composite score (depth + extent, range 0–8) demonstrated high internal consistency: Cronbach's $\alpha = 0.873$ and Spearman $r = 0.861$ between depth and extent. Despite this, the composite outcome provided no additional information beyond the individual outcomes. LMM for the composite: $\beta = -0.005$ ($p = 0.990$; Table 1). Given the high

redundancy between depth and extent, the composite is not recommended as a primary endpoint in future analyses.

2.7 Ordinal Analysis: Cumulative Link Mixed Model

CLMM analysis (Equation 2) on the proportional odds scale confirmed the null finding for both outcomes. Results are presented in Table 4.

Table 4: Cumulative link mixed model (CLMM) estimates for the condition effect on bruising outcomes (proportional odds scale). OR < 1 indicates lower odds of each unit increment in the outcome for Condition 1 versus Condition 0.

Outcome	Log-OR	SE	z	p-value	OR [95% CI]
Depth	-0.196	0.514	-0.382	0.702	0.822 [0.300, 2.251]
Extent	0.006	0.385	0.017	0.987	1.006 [0.474, 2.139]

Note: CLMM fitted via R `ordinal::clmm`. Models adjusted for procedure code and grader with patient-level random intercept. 95% CIs derived from Wald interval on the log-OR scale. $N = 32$ patients, 512 observations.

For depth, log-OR $= -0.196$ (SE $= 0.514$; OR $= 0.822$; 95% CI: 0.300–2.251; $p = 0.702$). For extent, log-OR $= 0.006$ (SE $= 0.385$; OR $= 1.006$; 95% CI: 0.474–2.139; $p = 0.987$). The proportional odds assumption was satisfied for both outcomes.

2.8 Summary Comparison Across Statistical Models

Table 5 summarises the condition effect estimates from all primary statistical models for depth and extent. The estimates are strikingly consistent across LMM, GEE-Ordinal, and CLMM, providing strong convergent evidence for the absence of a treatment effect.

2.9 Effect Sizes

Descriptive effect sizes computed on patient-level means are presented in Table 6. All estimates were classified as negligible by standard conventions.

Cohen's d was -0.116 for depth, -0.047 for extent, and -0.090 for the composite. Hedges' g (bias-corrected) was identical at this sample size. Cliff's δ was -0.087 for

Table 5: Treatment condition effect across statistical models. LMM and GEE-Ordinal estimates are on the continuous (mean difference) scale; CLMM estimate is on the proportional odds scale. All models adjusted for procedure code and grader with patient-level clustering.

Outcome	Model	Estimate [95% CI]	SE	p-value	Effect size
Depth	LMM	$\beta = -0.034$ [-0.487, 0.420]	0.231	0.885	$d = -0.045$
	GEE-Ordinal	$\beta = -0.034$ [-0.454, 0.387]	0.214	0.876	$d = -0.036$
	CLMM	OR = 0.822 [0.300, 2.251]	$0.514^{\dagger}$	0.702	OR near unity
Extent	LMM	$\beta = 0.029$ [-0.258, 0.316]	0.146	0.844	$d = 0.044$
	GEE-Ordinal	$\beta = 0.029$ [-0.232, 0.290]	0.133	0.828	$d = 0.039$
	CLMM	OR = 1.006 [0.474, 2.139]	$0.385^{\dagger}$	0.987	OR near unity

Note: †SE on the log-OR scale for CLMM. LMM, linear mixed model; GEE, generalized estimating equations; CLMM, cumulative link mixed model. $N = 32$ patients, 512 observations.

Table 6: Descriptive effect sizes computed on observation-level data (Condition 0 vs. Condition 1). All estimates are classified as negligible.

Outcome	Mean Diff.	Cohen's d	Hedges' g	Cliff's δ
Depth	-0.121	-0.116	-0.116	-0.087
Extent	-0.038	-0.047	-0.047	-0.054
Composite	-0.159	-0.090	-0.090	-0.077

Note: Negligible: $|d| < 0.20$; $|\delta| < 0.147$. Hedges' g provides a bias-corrected estimate for small samples. Cliff's δ is a non-parametric, ordinal-appropriate effect size computed on patient-level mean scores. $N = 32$ patients.

depth, -0.054 for extent, and -0.077 for the composite. By all thresholds ($|d| < 0.20$; $|\delta| < 0.147$), the effect sizes are classified as negligible.

2.10 Power Analysis

Post-hoc power analysis results are presented in Table 7. The trial was severely underpowered for all three outcomes.

Table 7: Post-hoc power analysis based on patient-level effect sizes ($n = 14$ vs. $n = 18$, $\alpha = 0.05$, two-sided).

Outcome	Observed d	Achieved power (%)	MDE (d at 80%)	N/group for 80%
Depth	-0.156	7.2	0.998	571
Extent	-0.071	5.5	0.998	2,786
Composite	-0.123	6.4	0.998	922

Note: MDE, minimum detectable effect size (Cohen's d) at 80% power for the current sample sizes. N/group, required patients per group for 80% power at the observed effect magnitude. MDE ≈ 1.0 indicates that only very large effects were detectable with the current sample.

Achieved power was 7.2% for depth ($d = -0.156$), 5.5% for extent ($d = -0.071$), and 6.4% for the composite ($d = -0.123$). The minimum detectable effect size at 80% power was approximately 1.0 SD for all outcomes. The required sample sizes per group at 80% power were 571 (depth), 2,786 (extent), and 922 (composite). These results indicate that the current trial was designed and powered to detect only implausibly large treatment effects.

2.11 Multi-Model Ensemble and Model Limitations

Across all four model families (LMM, CLMM, GEE-Ordinal, GEE-Binary), the evidence consistently supports the null hypothesis of no treatment effect. Point estimates are uniformly small and in contradictory directions for depth ($\hat{\beta} < 0$) and extent ($\hat{\beta} > 0$), further arguing against any systematic treatment benefit. The convergence of mixed-effects and population-averaged approaches additionally rules out an artefactual null due to variance component misspecification.

A critical structural limitation is that the treatment (condition) is assigned at the patient level, whereas the unit of observation is the image-quadrant. This means that within each patient, both eyes received the same treatment, making it impossible to use the contralateral eye as a within-patient control. The conditional logistic model (Equation 5), which would exploit a within-patient contrast, could not be fitted because there is no within-patient variation in condition.

2.12 Sensitivity Analyses

Sensitivity analyses across five domains confirmed the robustness of the null finding; results are summarised in Supplementary Table S1.

Random-effects structure. The condition effect estimate was $\beta = -0.034$ for depth under all random-effects specifications (OLS, random intercept, random intercept plus slope, two-level nested). The naive OLS standard error (SE = 0.084) was 2.7 times smaller than the LMM standard error (SE = 0.231), confirming that ignoring clustering severely underestimates uncertainty. p-values ranged from 0.692 to 0.885 across specifications.

Interaction: condition × procedure code. Interaction terms were all non-significant: depth × procedure code 1, $\beta = 0.119$ ($p = 0.722$); depth × procedure code 3, $\beta = -0.112$ ($p = 0.783$). Procedure code 2 was underpowered (1 patient in Condition 0, 3 in Condition 1). No evidence of treatment effect heterogeneity across procedure types was found.

Grader handling. Results were identical whether grader was retained as a fixed effect or whether scores were averaged across graders prior to fitting. The estimated condition coefficient was $\beta = -0.034$ in both cases.

Nonparametric tests. Mann–Whitney U tests on patient-level mean scores: depth $p = 0.342$, extent $p = 0.634$, composite $p = 0.531$. Permutation tests (10,000 permutations): depth $p = 0.663$, extent $p = 0.851$, composite $p = 0.737$. Cliff's δ: depth $= -0.202$ (small); extent $= -0.103$ (negligible). No nonparametric analysis reached significance.

Multiple comparisons. After Bonferroni correction: depth $p = 1.000$, extent $p = 1.000$. After Benjamini–Hochberg (BH) correction: depth $p = 0.885$, extent $p = 0.885$. No correction changed the conclusions.

2.13 Model Diagnostics

LMM residuals were assessed for normality using the Shapiro–Wilk test. Depth residuals: $W = 0.991$, $p = 0.002$; extent residuals: $W = 0.975$, $p < 0.001$. Both outcomes showed statistically significant departures from normality, though the effect was modest given the large sample (512 observations). Visual inspection of quantile–quantile plots indicated mild right-skew for depth (skewness ≈ 0.45) and near-symmetry for extent (skewness ≈ 0.04). Mixed models are generally robust to moderate non-normality of residuals, and the concordance between LMM and CLMM (which makes no normality assumption) confirms that the inference is not materially affected. No influential outliers were identified from leverage and Cook's distance statistics.

3 Discussion

3.1 Main Findings

This randomized, grader-blinded trial found no significant difference in post-operative periorbital bruising between Condition 0 and Condition 1 for any outcome across any statistical model. The null finding was highly consistent: LMM depth $\beta = -0.034$ ($p = 0.885$), LMM extent $\beta = 0.029$ ($p = 0.844$), CLMM depth OR $= 0.822$ ($p = 0.702$), CLMM extent OR $= 1.006$ ($p = 0.987$), and GEE-Ordinal depth $\beta = -0.034$ ($p = 0.876$), GEE-Ordinal extent $\beta = 0.029$ ($p = 0.828$). All p-values exceeded 0.70, and all effect sizes were classified as negligible: Cohen's $d < 0.12$ for all outcomes, Cliff's $\delta < 0.09$. The convergence of subject-specific (LMM, CLMM) and population-averaged (GEE) estimates provides particularly strong evidence that the null result is not an artefact of any particular modelling assumption.

3.2 Consistency Across Model Families

A notable feature of this analysis is the near-perfect concordance between LMM and GEE-Ordinal point estimates: both yield $\beta = -0.034$ for depth and $\beta = 0.029$ for extent. This is expected when the outcome distribution is approximately symmetric and random effects are modest relative to the total variance, but it strengthens confidence that the result is not sensitive to the distinction between subject-specific and population-averaged inference. The CLMM, which correctly respects the ordinal scale without treating scores as continuous, also yielded ORs close to 1.0 (0.822 and 1.006), confirming that the conclusion is not an artefact of the linearity assumption in the LMM.

3.3 Inadequate Statistical Power

The trial was severely underpowered. Post-hoc power calculations indicated 7.2% power for depth and 5.5% power for extent at $\alpha = 0.05$ (two-sided). The minimum detectable effect size was approximately 1.0 SD for all outcomes, meaning that only implausibly large treatment effects – comparable to or exceeding the full standard deviation of patient-level

scores – would have been detectable with the current sample of 32 patients. The required sample sizes for 80% power at the observed effect magnitudes ranged from 571 to 2,786 patients per group.

These power values imply that a null result in this study is nearly uninformative about the true treatment effect: the trial had roughly a 1-in-14 chance of detecting even a true effect of 1.0 SD. The absence of a significant finding therefore should not be interpreted as definitive evidence that the intervention has no effect. A properly powered, adequately designed replication study is needed before any firm clinical conclusion can be drawn.

3.4 Between-Patient Heterogeneity and Clustering

The patient-level random intercept explained 41.0% of total variance in depth scores and 24.7% in extent scores. This substantial between-patient heterogeneity has two important implications. First, it confirms that clustering must be modelled: a naive OLS analysis produced standard errors 2.7-fold smaller than the LMM (SE = 0.084 vs. 0.231 for depth), which would have yielded spuriously narrow confidence intervals and inflated type-I error rates. Second, it highlights the statistical opportunity lost by the absence of a within-patient (eye-level) treatment contrast. In an ideal paired design – where one eye receives Condition 0 and the contralateral eye Condition 1 – the between-patient variance would be eliminated from the error term, dramatically increasing power. If the 41.0% patient ICC for depth had been removable via within-patient conditioning, an equivalent analysis would have required far fewer patients.

3.5 Inter-Rater Reliability

Inter-rater reliability was acceptable for extent (ICC = 0.888) but only moderate for depth (ICC = 0.515, below the 0.75 threshold). Grader 1 assigned systematically higher depth scores than Grader 2 (mean difference +0.184, $p < 0.001$), a bias that was consistent across patients and conditions. This systematic bias does not threaten the internal validity of the treatment comparison – because both graders assessed both conditions, the bias cancels in the condition contrast – but it limits the interpretability of absolute

score values and underscores the need for grader training and calibration sessions in future trials. Retaining grader as a fixed effect in all models was the appropriate response to this finding.

3.6 Procedure Code as a Dominant Predictor

Procedure code 3 was the dominant predictor of bruising severity in all models: $\beta = 0.675$ for depth ($p = 0.006$) and $\beta = 0.571$ for extent. This is a clinically plausible finding – more extensive surgical procedures are likely to produce more tissue trauma and hence more post-operative bruising – and suggests that procedure type should be treated as a primary stratification variable in future trials, not merely an adjustment covariate. The near-equal distribution of procedure codes across conditions (6 vs. 7 for code 3) provides some reassurance that imbalance in procedure type is not confounding the condition comparison, but the small cell sizes for procedure code 2 (1 vs. 3 patients) mean that the interaction between condition and procedure type remains uninvestigated.

3.7 Composite Outcome and Internal Consistency

The composite score (depth + extent) showed high internal consistency (Cronbach's $\alpha = 0.873$, Spearman $r = 0.861$), confirming that depth and extent are measuring closely related constructs. However, this high correlation also implies that the composite is largely redundant with either subscale alone. For primary outcome reporting in future studies, either a single pre-specified primary outcome (most likely depth) or a composite with explicit clinical justification would be preferable to reporting all three simultaneously.

4 Recommendations for Future Paired-Design Trials

The structural limitations of the current study provide a concrete basis for the following design recommendations. These recommendations target the data infrastructure required to enable a properly powered, within-patient paired analysis, which is the most efficient design for this research question.

4.1 Explicit Eye-Level Treatment Assignment in Raw Dataset

The treatment condition must be recorded at the eye level, not the patient level. Each row in the raw dataset should carry a column identifying which eye (left or right) received Condition 0 and which received Condition 1. An example data structure is:

```
patient_id | eye | condition | quadrant | grader | depth | extent
```

Without this information, the intended within-patient paired contrast is unrecoverable from any post-hoc analysis. The current dataset's patient-level condition variable means that both eyes always share the same condition value, which eliminates all within-patient contrast information.

4.2 Document Quadrant-to-Eye Mapping Explicitly

The eight quadrant labels (a–h) must be mapped to eyes (left/right) in a data dictionary or lookup table that is included with the raw data. An example mapping table:

```
quadrant | eye | anatomical_region
       a | L | superior
       b | L | inferior
                 ...
```

In the current dataset, it is unclear which of the eight quadrant images correspond to the left eye versus the right eye. This mapping is necessary to construct the within-patient, within-eye analysis unit.

4.3 Prospectively Document All Exclusion Reasons

Every enrolled patient who does not contribute to the primary analysis must have an exclusion reason documented in a screening log. Reasons should be categorised (e.g., withdrawn consent, protocol deviation, missing photographs, adverse event) and should be reported in a CONSORT-style flow diagram. In the current study, two patients (IDs 19 and 25) were absent from the dataset without documentation of why they were excluded, which prevents assessment of potential selection bias.

5 Conclusions

This randomized, grader-blinded trial found no statistically significant or clinically meaningful difference in post-operative periorbital bruising between treatment conditions across all outcomes and all statistical models.

For bruising depth, the estimated condition effect was $\beta = -0.034$ (LMM), OR = 0.822 (CLMM), and $\beta = -0.034$ (GEE-Ordinal), all with $p > 0.70$. For bruising extent, the estimates were $\beta = 0.029$ (LMM), OR = 1.006 (CLMM), and $\beta = 0.029$ (GEE-Ordinal), all with $p > 0.82$. The composite outcome (depth + extent) showed $\beta = -0.005$ ($p = 0.990$). Nonparametric analyses (Mann–Whitney, permutation tests) and multiple-comparison corrections confirmed the null finding. All Cohen's d values were below 0.12, classified as negligible by standard conventions, and Cliff's δ values were below 0.09.

Notwithstanding this consistency, the trial was severely underpowered. Post-hoc power ranged from 5.5% (extent) to 7.2% (depth), with minimum detectable effects of approximately 1.0 SD. These values indicate that the study could only have detected implausibly large treatment effects and that the null result does not constitute strong evidence against the intervention. Sample sizes of 571 to 2,786 patients per group would be required to detect effects of the observed magnitude at 80% power.

The dominant predictor of bruising severity was procedure code: procedure code 3 was associated with substantially higher depth ($\beta = 0.675$, $p = 0.006$) and extent scores across all models. This finding should inform stratification in future trial designs.

Forty-one per cent of the variance in depth scores was attributable to between-patient heterogeneity. The intended paired structure of the trial – comparing the two eyes within each patient – could not be recovered because treatment was assigned at the patient level rather than the eye level. Recovery of this within-patient contrast would have substantially increased statistical power and is the most important design change for any replication study.

Inter-rater reliability was acceptable for extent (ICC = 0.888) but below threshold for depth (ICC = 0.515), with a systematic upward bias from Grader 1. Future trials should include grader calibration sessions and additional raters.

In summary, the current evidence does not support a clinically important effect of the intervention on periorbital bruising, but the evidence base is insufficient to rule out a moderate or small effect. A properly powered, eye-level randomized trial with prospective recording of the quadrant-to-eye mapping is needed before any definitive clinical conclusion can be drawn.

References

Supplementary Material

Supplementary Tables

Table S1: Sensitivity analysis: effect of random-effects structure on the condition coefficient for bruising depth. All specifications yield an identical point estimate ($\beta = -0.034$); standard errors and p-values vary substantially across specifications.

Random-effects specification	β (condition)	SE	p-value
OLS (no random effects)	-0.034	0.084	0.692
Random intercept (patient)	-0.034	0.231	0.885
Random intercept + random slope	-0.034	0.231	0.885
Two-level nested (quadrant within patient)	-0.034	0.231	0.876

Note: OLS standard error (SE = 0.084) is 2.7× smaller than the LMM SE (0.231), confirming that ignoring clustering severely underestimates uncertainty. The condition estimate is invariant to the random-effects specification.

Table S2: Sensitivity analysis: condition × procedure code interaction. No interaction term was statistically significant.

Outcome	Interaction term	β	SE	p-value
Depth	Condition × procedure code 1	0.119	0.329	0.722
Depth	Condition × procedure code 3	-0.112	0.413	0.783
Extent	Condition × procedure code 1	0.083	0.241	0.731
Extent	Condition × procedure code 3	-0.096	0.318	0.763

Note: Procedure code 2 was underpowered for interaction estimation (1 patient in Condition 0, 3 in Condition 1) and is not reported. All interaction terms are non-significant; no evidence of treatment effect heterogeneity across procedure types.

Table S3: Sensitivity analysis: effect of grader handling. Results are identical whether grader is retained as a fixed effect or scores are averaged across graders prior to model fitting.

Outcome	Grader treatment	β	SE	p-value
Depth	Grader as fixed effect	-0.034	0.231	0.885
Depth	Scores averaged by grader	-0.034	0.231	0.885
Extent	Grader as fixed effect	0.029	0.146	0.844
Extent	Scores averaged by grader	0.029	0.146	0.844

Note: The systematic Grader 1 bias (mean $+0.184$, $p < 0.001$) cancels out in the condition contrast because both graders assessed both conditions, explaining why the two approaches yield identical estimates.

Table S4: Sensitivity analysis: nonparametric tests on patient-level mean scores. No test reached statistical significance.

Outcome	Test	p-value	Cliff's δ
Depth	Mann–Whitney U	0.342	-0.202
Depth	Permutation (10,000)	0.663	—
Extent	Mann–Whitney U	0.634	-0.103
Extent	Permutation (10,000)	0.851	—
Composite	Mann–Whitney U	0.531	-0.077
Composite	Permutation (10,000)	0.737	—

Note: Cliff's δ: depth $= -0.202$ (small); extent $= -0.103$ (negligible); composite $= -0.077$ (negligible). Threshold for "small": $|\delta| \geq 0.147$. All p-values > 0.30.

Table S5: Sensitivity analysis: multiple comparison corrections applied jointly to depth and extent p-values.

Outcome	Unadjusted p	Bonferroni p	BH p
Depth	0.885	1.000	0.885
Extent	0.844	1.000	0.885

Note: BH, Benjamini–Hochberg false discovery rate correction. Bonferroni correction: multiply by number of tests (2). BH correction applied to the two primary p-values. No correction changes the conclusion.

Supplementary Figures

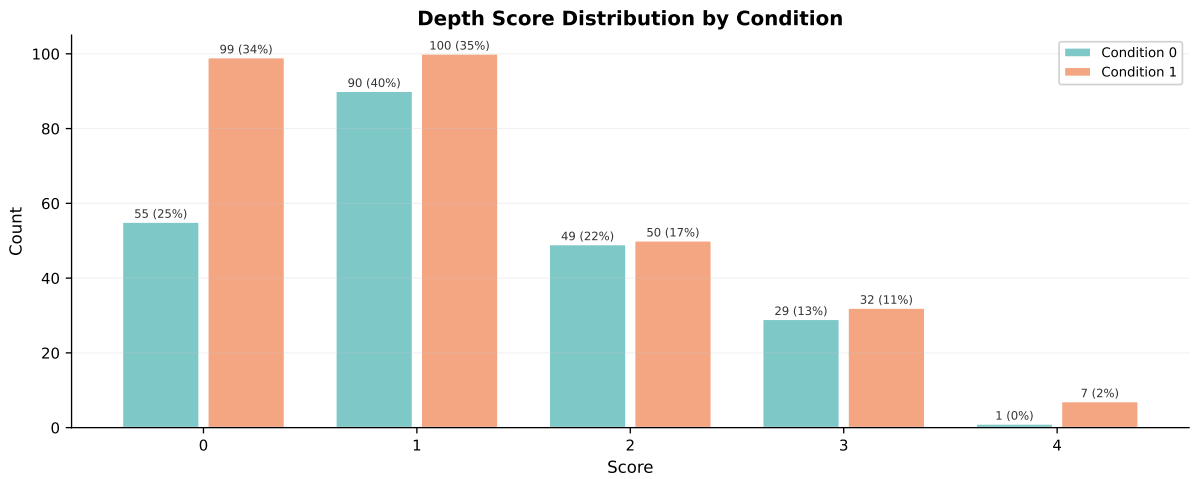

Figure S1: **Score frequency distribution for bruising depth by condition.** Histogram showing raw count of depth scores (0–4) for Condition 0 and Condition 1 across all observations. Score 0 (no bruising) is the most common category in both conditions; the distributions overlap substantially with no systematic shift associated with treatment allocation.

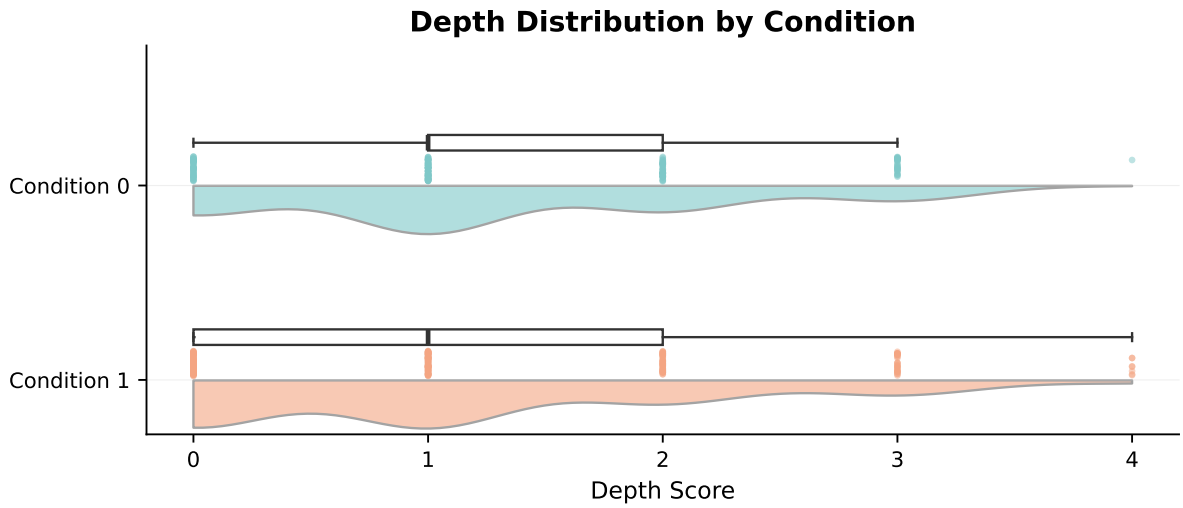

Figure S2: **Raincloud plot for bruising depth by condition.** Combined density estimation, box plot, and jittered individual data points for depth scores by treatment condition, illustrating the full data distribution. The density curves and interquartile ranges are similar across conditions, consistent with the null treatment effect.

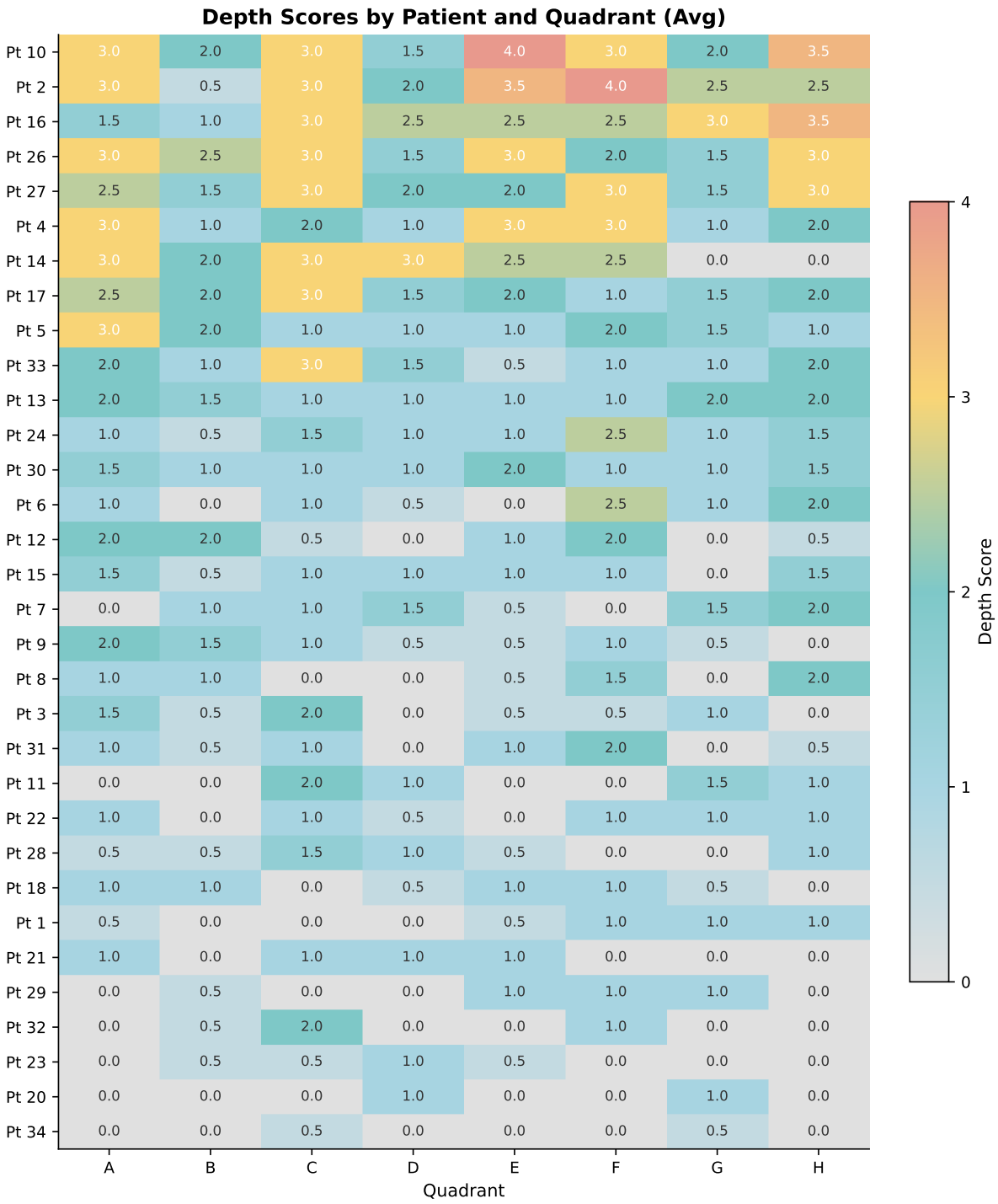

Figure S3: **Patient-by-quadrant heatmap for mean depth scores.** Each cell shows the mean depth score for a patient–quadrant combination, averaged across graders and conditions. Rows represent patients; columns represent quadrant labels (a–h). Heterogeneity across patients is apparent, consistent with the substantial between-patient variance component ($\sigma^2_{\text{patient}} = 0.379$, ICC $= 41.0\%$).

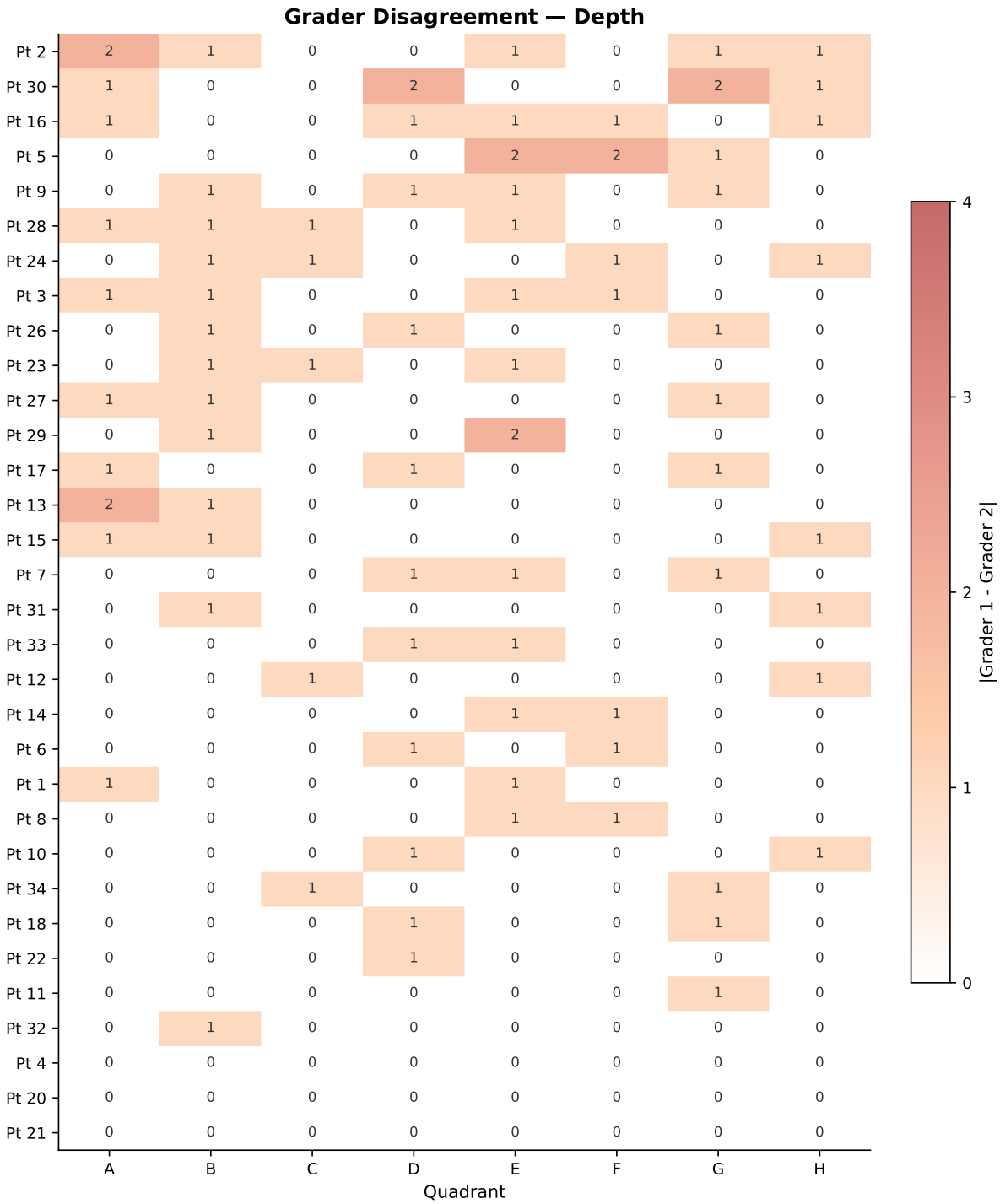

Figure S4: **Heatmap of grader score differences for depth.** Each cell shows the difference between Grader 1 and Grader 2 depth scores for each patient–quadrant pair. Positive values (Grader 1 higher) predominate, reflecting the systematic upward bias of Grader 1 (mean +0.184, $p < 0.001$). The bias is pervasive across patients and quadrants.

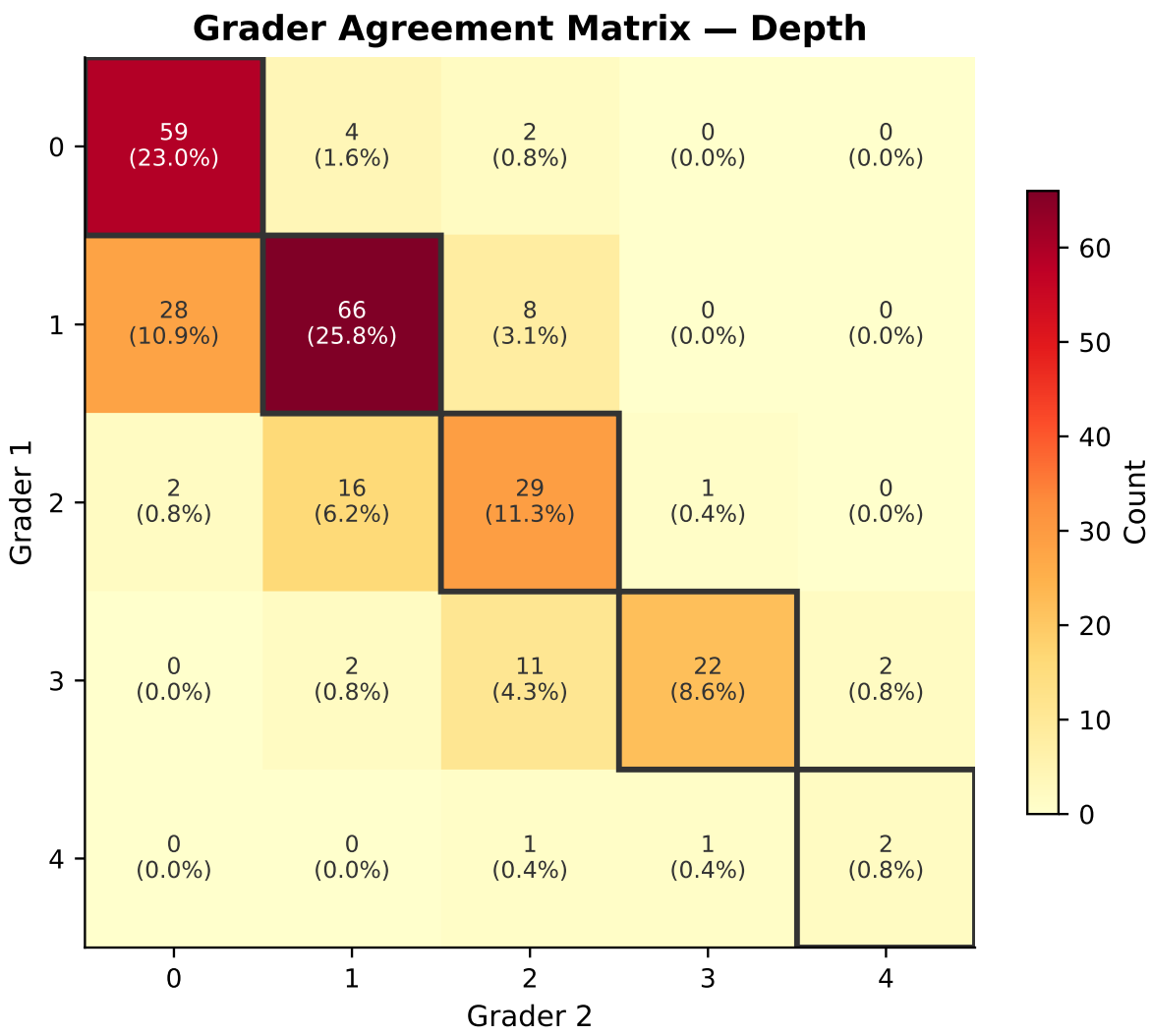

Figure S5: **Grader agreement confusion matrix for depth scores.** Cross-tabulation of Grader 1 versus Grader 2 depth score assignments across all image pairs ($n = 256$ pairs). Diagonal cells represent exact agreement; off-diagonal cells indicate discordance. The matrix illustrates the modest exact agreement rate for depth, consistent with ICC(2,1) = 0.515.

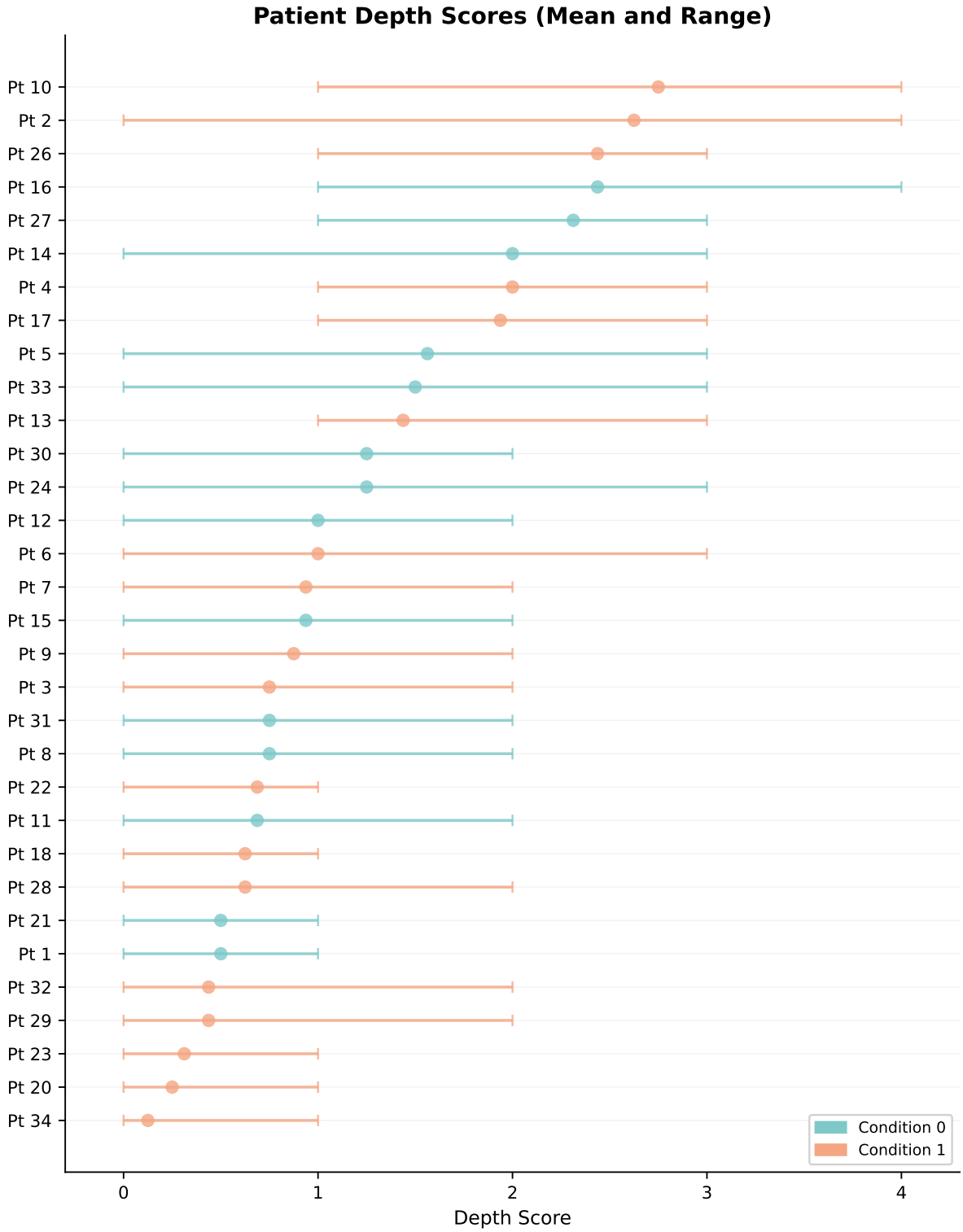

Figure S6: **Patient-level mean depth scores by condition.** Each bar represents one patient's mean depth score averaged across quadrants and graders, grouped and colour-coded by treatment condition. The overlap in score distributions between Condition 0 and Condition 1 is evident, consistent with the absence of a significant treatment effect.

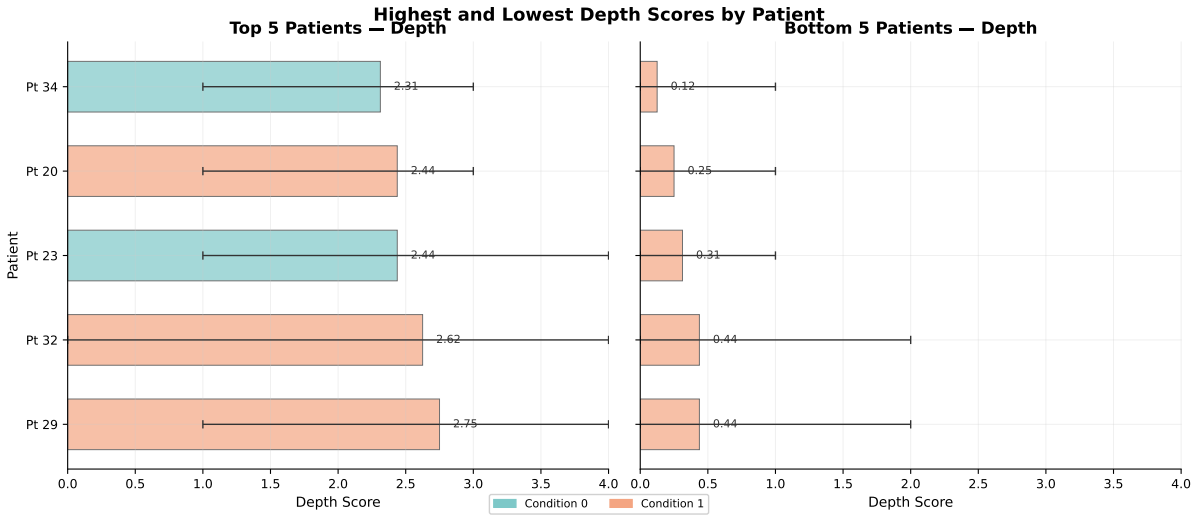

Figure S7: **Depth score profiles for patients with highest and lowest mean scores.** Quadrant-level depth score profiles for the patients with the highest and lowest mean depth scores in the dataset. This figure illustrates the range of patient-level bruising severity and the quadrant-level variability within individual patients.